

What Measures to Choose for Effective Pattern Evaluation?

- ❑ Null value cases are predominant in many large datasets
 - ❑ Neither milk nor coffee is in most of the baskets; neither Mike nor Jim is an author in most of the papers;
- ❑ *Null-invariance* is an important property
- ❑ Lift, χ^2 and cosine are good measures if null transactions are not predominant
 - ❑ Otherwise, *Kulczynski + Imbalance Ratio* should be used to judge the interestingness of a pattern
- ❑ Exercise: Mining research collaborations from research bibliographic data
 - ❑ Find a group of frequent collaborators from research bibliographic data (e.g., DBLP)
 - ❑ Can you find the likely advisor-advisee relationship and during which years such a relationship happened?
 - ❑ Ref.: C. Wang, J. Han, Y. Jia, J. Tang, D. Zhang, Y. Yu, and J. Guo, "Mining Advisor-Advisee Relationships from Research Publication Networks", KDD'10